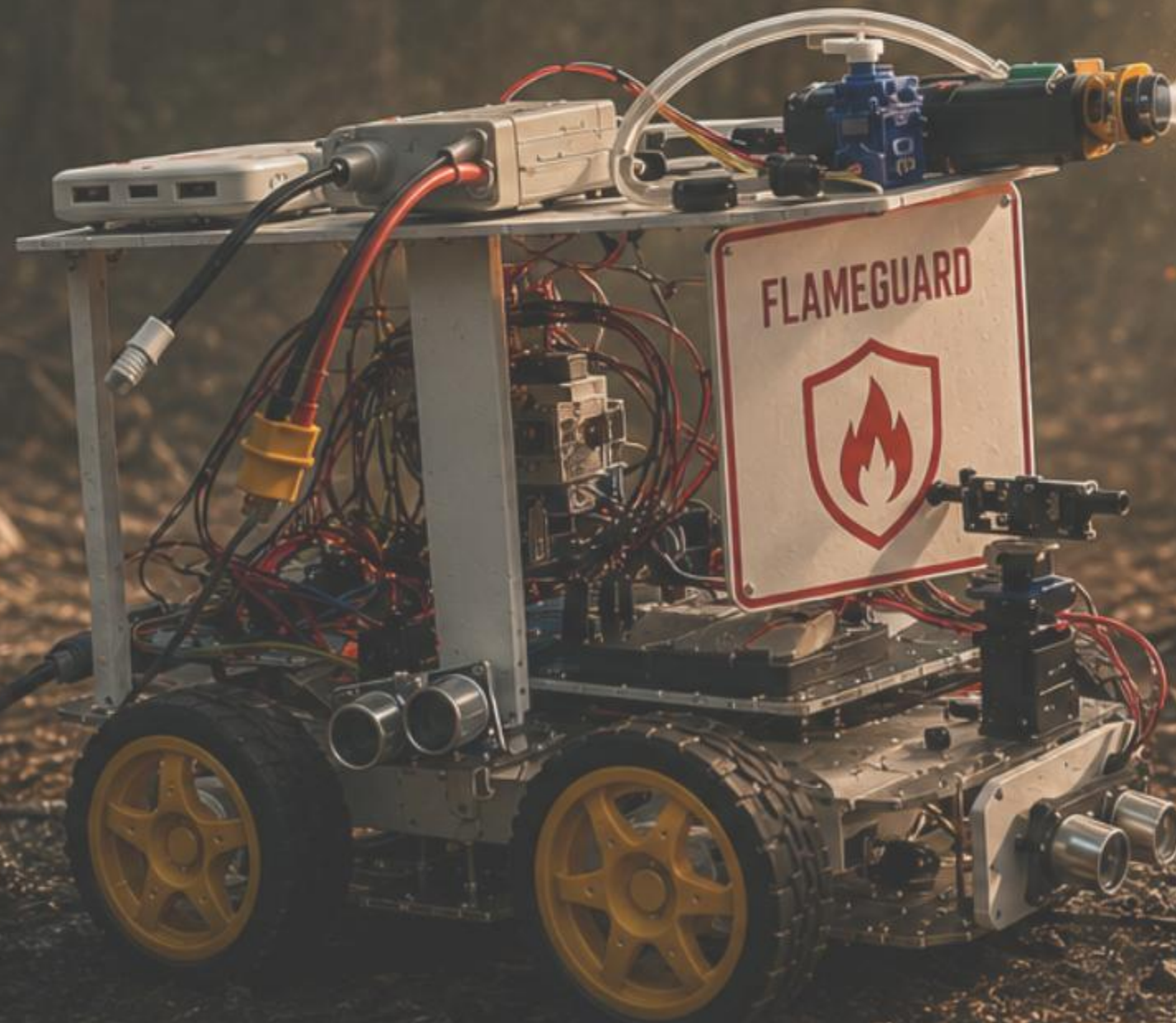


FLAMEGUARD

FIREFIGHTING ROBOT

CSE 437 — Robotics Lab | University of Asia Pacific

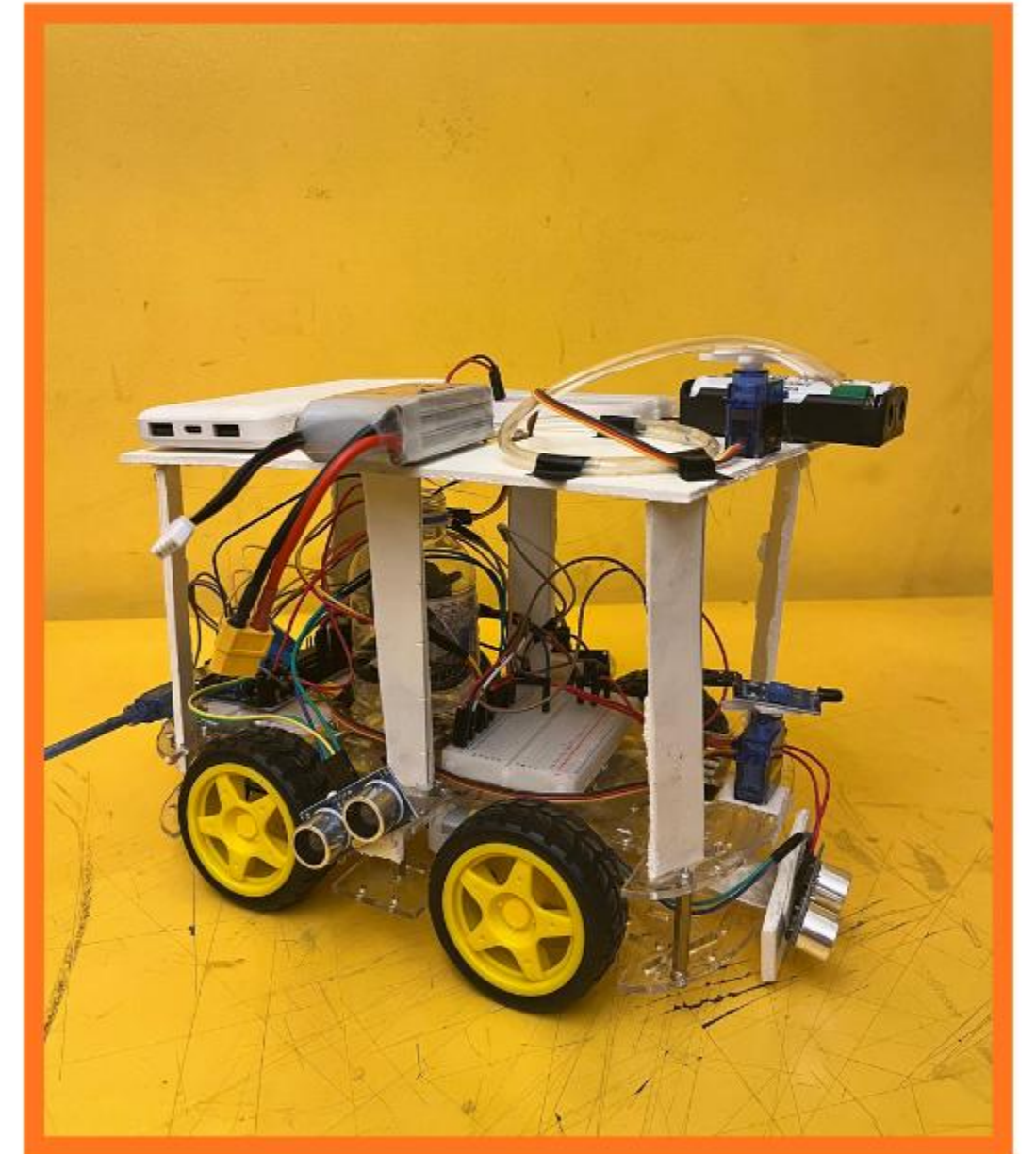


Presented By :
Nusrat Jahan Urmi
ID : 22201214

Presented To :
Nazmun Nahid
Assistant Professor
Department of CSE
University of Asia Pacific

Executive Summary

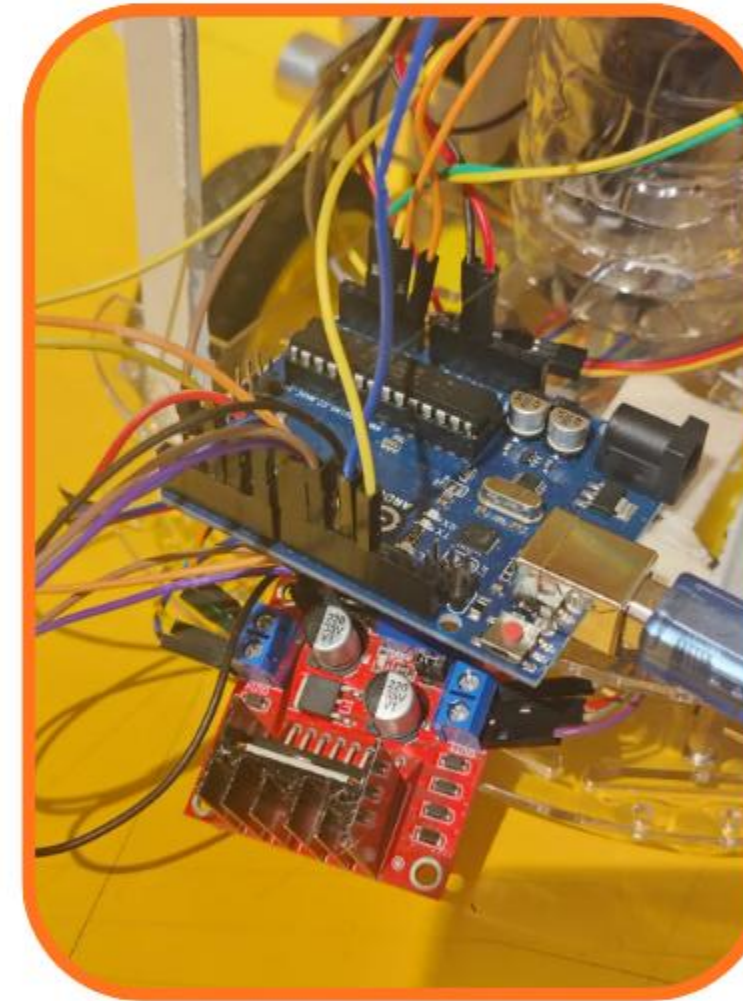
- 1  Automated system designed to detect and extinguish fire
- 2  Uses flame sensors to identify fire sources automatically
- 3  Moves toward the fire using a motor-driven mechanism
- 4  Water pump system sprays water to control and extinguish fire
- 5  Built using Arduino Uno, sensors, motors, and a servo motor



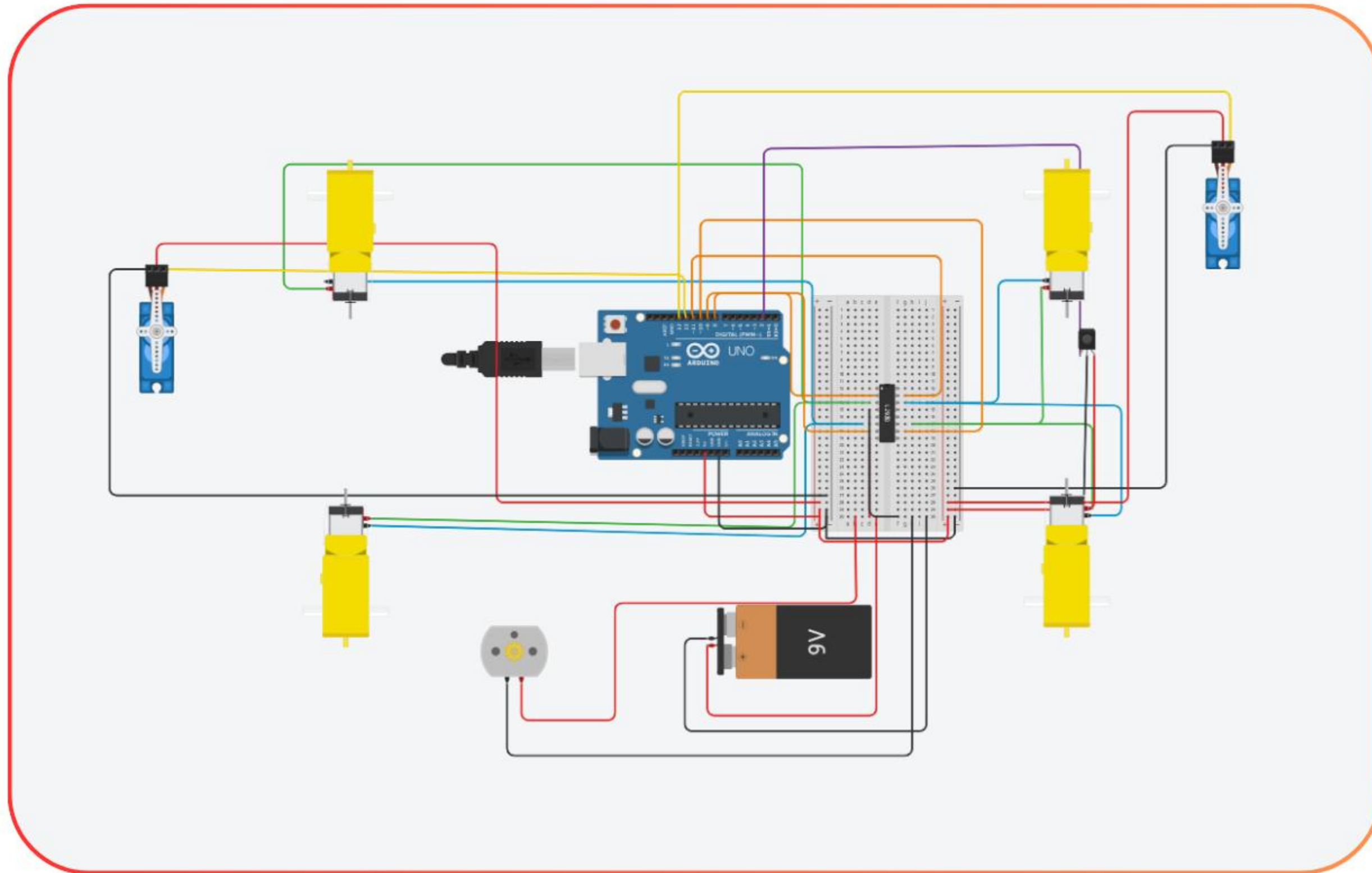
Hardware Contribution

Mechanical & Movement Hardware

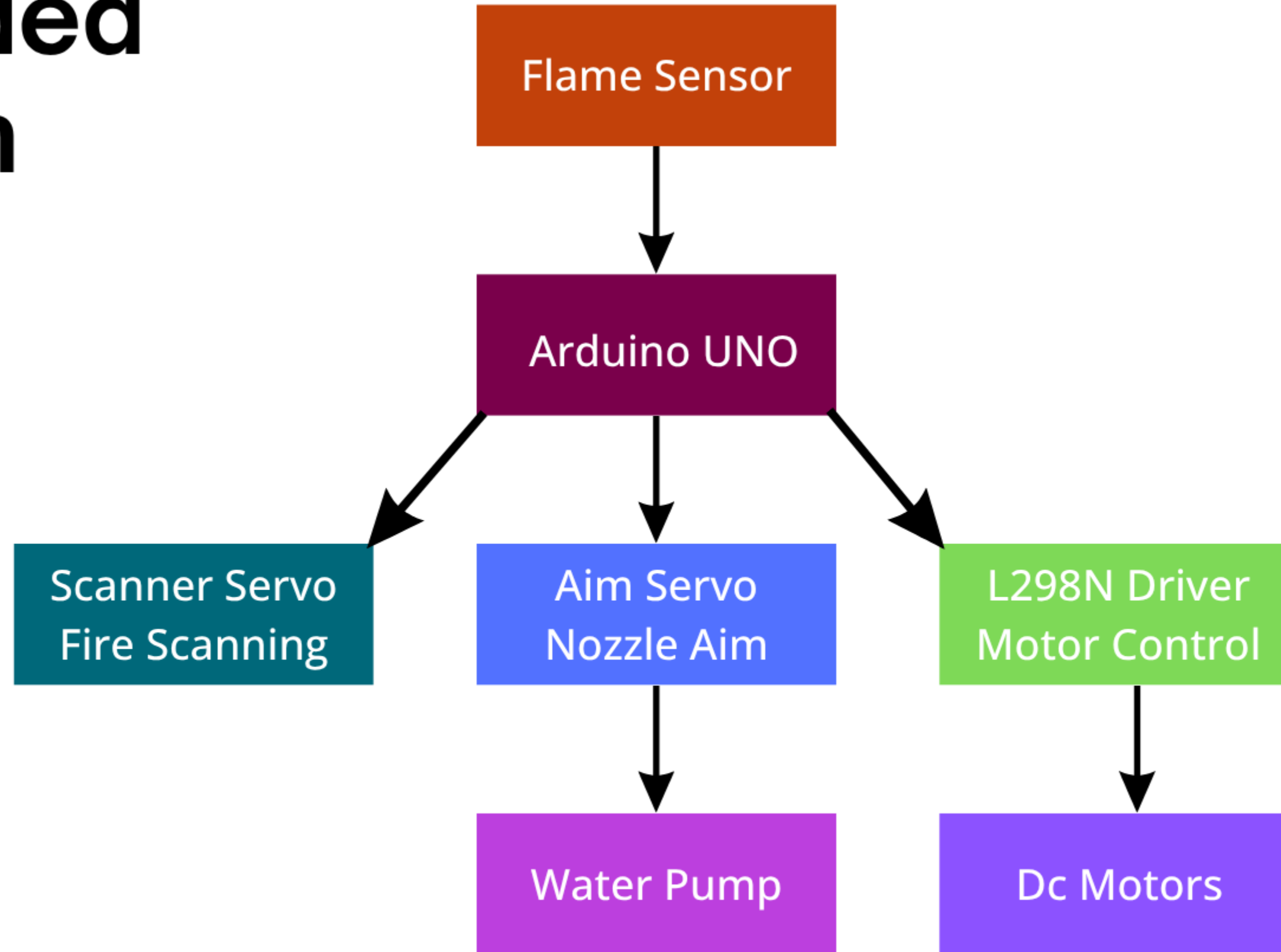
- 4WD chassis assembly
- BO motor fitting
- Wheel setup
- L298N motor driver connection
- Motor wiring
- Power supply connection for movement
- Robot movement testing
- Forward, backward, left, right movement checking
- Ultrasonic sensor placement
- Tested all individual robot movement functions.



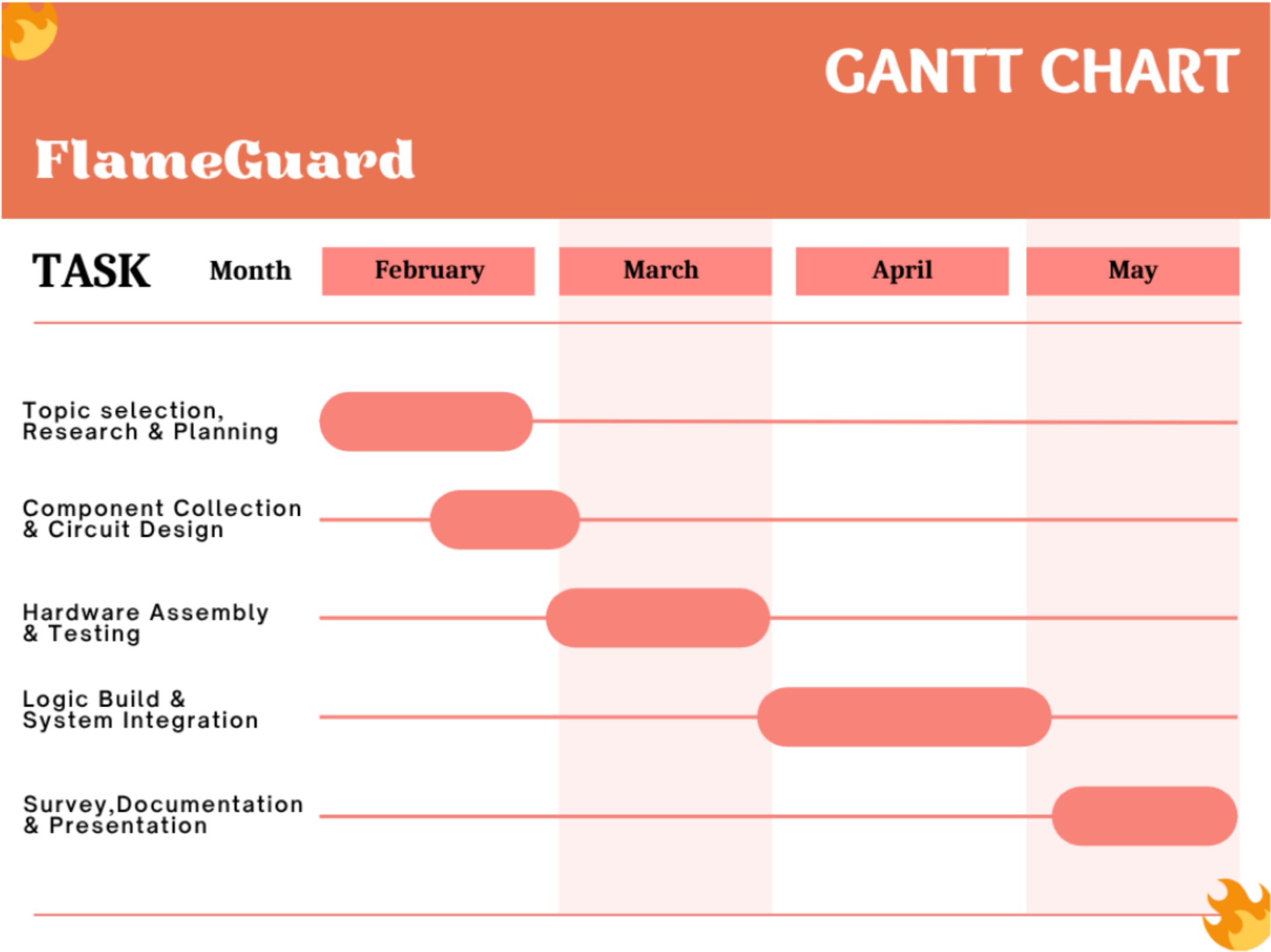
Circuit Diagram



Embedded Diagram

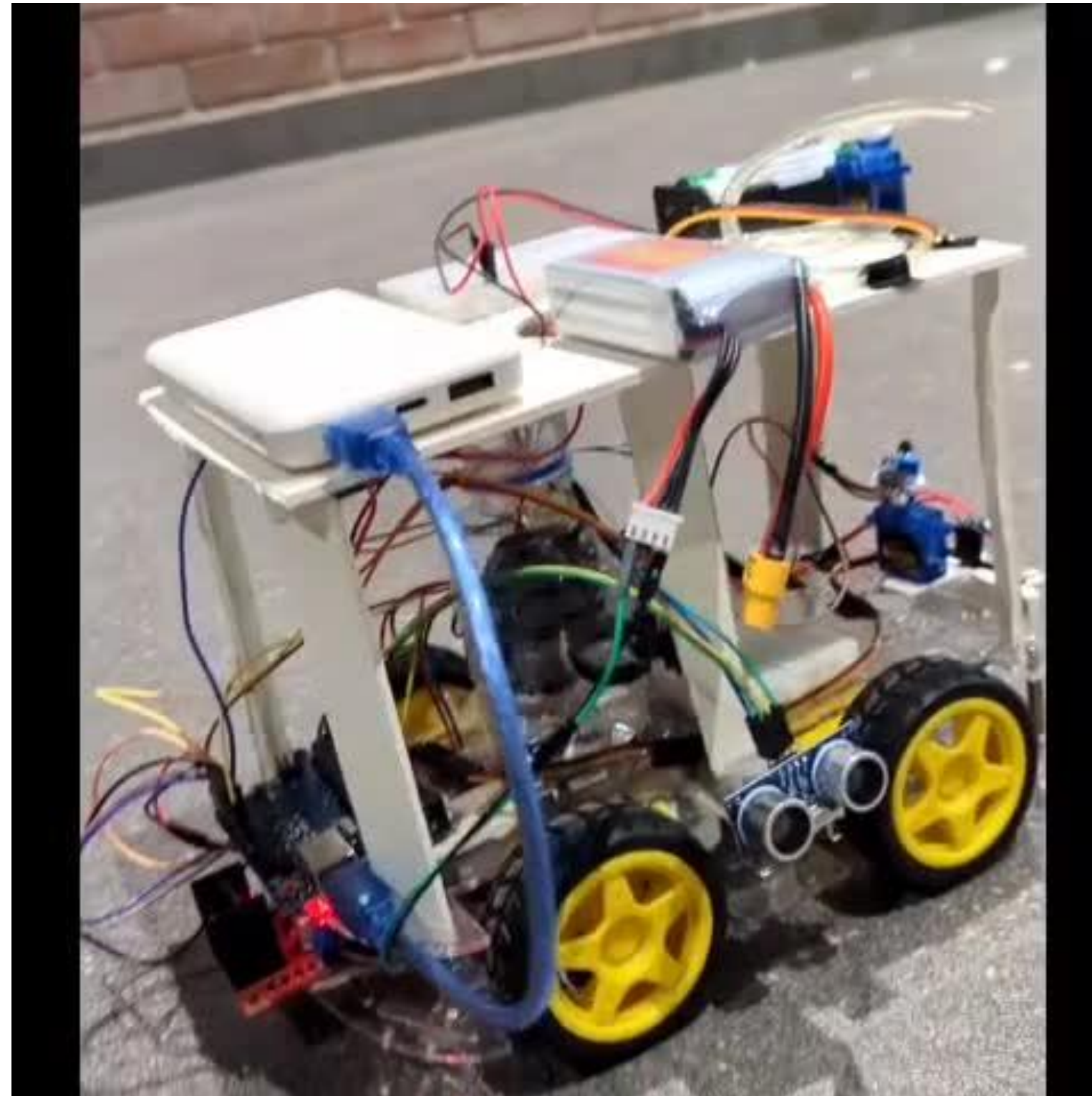


Gantt Chart

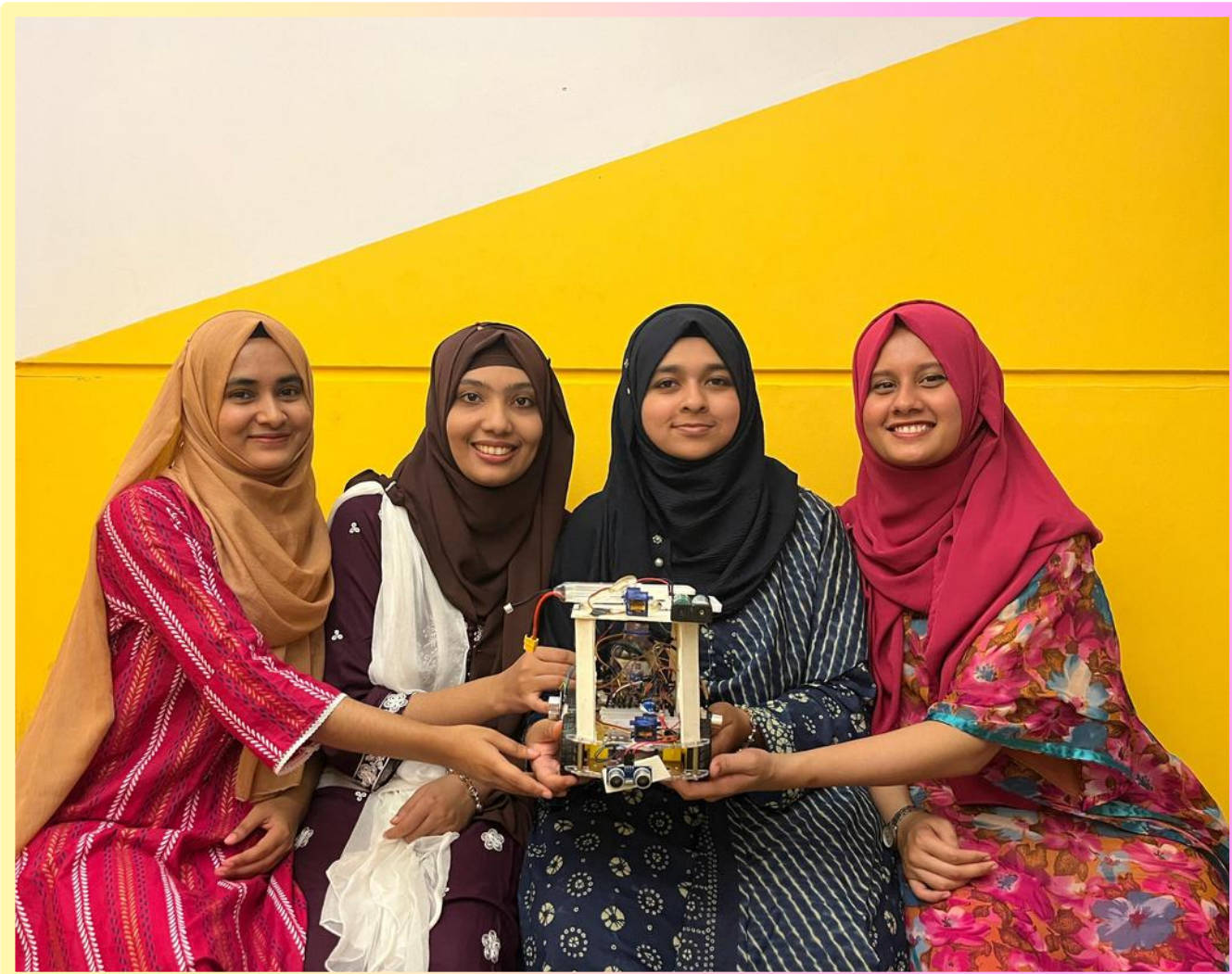


FlameGuard

Robot Demo Video



FlameGuard



Thank You

Any Question?